PIYUSH RAI

Full Stack Developer | AI & Data Science
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SUMMARY

Full Stack Developer with a B.Tech in Artificial Intelligence and Data Science, currently building multi-school EdTech products at Code and Learn Edutech. Experienced with Django, React, TypeScript, MS SQL Server, Celery and IIS deployment, plus machine learning model development and Flask API integration.

EXPERIENCE

Full Stack Developer | Code and Learn Edutech Private Limited | Sep 2025 – Present

- Developed the school_access Django application to support a multi-school architecture; wrote complex queries and managed schemas in MS SQL Server using SSMS.
- Engineered the "Master Academic Plan" logic to automate schedule distribution across schools.
- Refactored the React and TypeScript frontend by centralizing API calls in bridgeService.ts, enabling dynamic data loading for Curriculum and Assignment pages.
- Extended the sc_ai_corrector to compile subject diagrams with LaTeX, RDKit, Matplotlib, Asymptote and Ghostscript.
- Resolved concurrency bottlenecks in Celery workers running multi-stage rendering pipelines, stabilizing the backend.
- Deployed the application to production on IIS; built and maintained the Admin Dashboard with endpoints that sync real-time data between the school database and UI.

Machine Learning Trainee Intern | IBM, CSRBOX | 2 months

- Built and deployed a heart disease prediction model on IBM Cloud for patient risk assessment.
- Trained and evaluated SVM and Logistic Regression models; served real-time predictions through a Flask API with Pickle model serialization.

PROJECTS

Adaptive Learning Platform | Hackathon Project

- Built an AI smart-classroom platform that generates lecture summaries and MCQs from lecture videos using GPT models and speech-to-text APIs.
- Designed teacher and student logins with face-recognition attendance, classroom management, a date-wise lecture dashboard and gamified badges for quizzes and blogs.

Lexical Insight Engine | Course Project

- Developed a Python pipeline that scrapes article text from URLs with Selenium, cleans it and applies NLP techniques to generate text-analysis insights.

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, SQL, C, HTML, CSS
Frameworks & Libraries: Django, Flask, Express, React.js, Celery, Pandas, NumPy, Matplotlib, RDKit, BeautifulSoup, Selenium
Databases: MS SQL Server (SSMS), MongoDB
Tools & Platforms: Git, GitHub, IIS, IBM Cloud, LaTeX, Figma
Core Concepts: Data Structures and Algorithms, OOP, Machine Learning, NLP, REST APIs

EDUCATION

B.Tech, Artificial Intelligence and Data Science | Samrat Ashok Technological Institute, Vidisha | 2021 – 2025 | CGPA: 7.50/10

Class XII | Saket Shishu Ranjan Higher Secondary School, Vidisha | 2021 | 89%

Class X | Saket Shishu Ranjan Higher Secondary School, Vidisha | 2019 | 91%

CERTIFICATIONS & ACHIEVEMENTS

- IBM: Machine Learning with Python; Data Analysis with Python; Data Visualization with Python; AI Fundamentals
- MongoDB: The Document Model
- Secured 5th place at HackHound 2.0 (24-hour hackathon), SRM University
- Participated in Kriyeta 3.0 Hackathon, Indore